

6.1 & 6.2 INTRODUCTION TO CHARACTERISTICS

Exam Questions Addressed:

- Define Method of Characteristics (MOC). [Multiple Years]
- Define characteristic curve. [Multiple Years]
- What is meant by MOC and why is it necessary? / Advantages of MOC. [2076 Bhadra, 2075 Bhadra]

6.1 Concept & Principle of MOC

Method of Characteristics (MOC):

The Method of Characteristics is a powerful mathematical technique used to solve partial differential equations (PDEs) by converting them into ordinary differential equations (ODEs). These resulting ODEs apply strictly along specific continuous paths (or lines) in the space-time domain, which can then be solved using numerical or graphical techniques.

Characteristic Curves:

Characteristics are physical curves plotted in the $x - t$ plane defined by the equation $\frac{dx}{dt} = V \pm c$, where V is the velocity of flow and c is the celerity of the wave. The term $V \pm c$ physically represents the absolute velocity at which a disturbance (or pressure wave) propagates through the fluid medium.

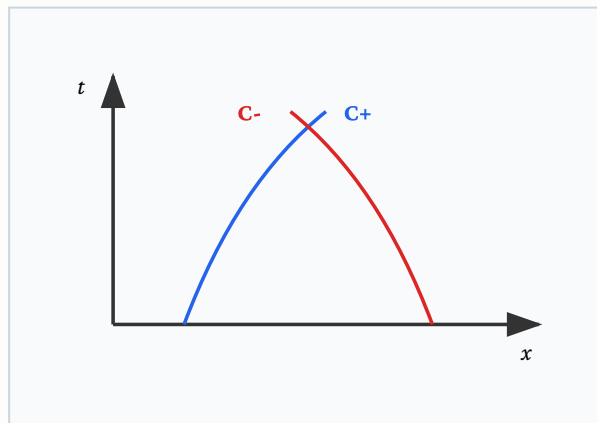


Figure: Characteristic Curves in $x-t$ plane

The slope of these characteristic lines is given by $\frac{dt}{dx} = \frac{1}{V \pm c}$.

Necessity and Advantages of MOC:

- **Simplification:** It mathematically transforms complex, coupled PDEs (like the Navier-Stokes equations for pipe flow) into easily integrable ODEs.
- **Accuracy:** It perfectly tracks the propagation of sharp wave fronts (like shock waves or water hammer), avoiding the numerical diffusion/dispersion seen in standard finite difference methods.
- **Boundary Constraints:** It conceptually isolates physical boundaries, allowing easy and accurate integration of complex boundary conditions (pumps, valves, reservoirs).



Exam Questions Addressed:

- Develop the characteristic equations from the partial differential form of unsteady pipe flow. [Multiple Years]
- Derive the expressions for the application of MOC to unsteady pipe flow problems. [2080 Chaitra]
- Develop a finite-difference solution of unsteady flow equations in terms of velocity and pressure / head and discharge. [Multiple Years]

6.4 Development of Characteristic Equations

The two basic PDEs governing 1D unsteady pipe flow are the Momentum and Continuity equations.

1. Momentum Equation:

$$L_1 = \frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} + \frac{1}{\rho} \frac{\partial P}{\partial x} + g \sin \theta + \frac{fV|V|}{2D} = 0$$

2. Continuity Equation:

$$L_2 = \frac{\partial P}{\partial t} + V \frac{\partial P}{\partial x} + \rho c^2 \frac{\partial V}{\partial x} = 0$$

Where: V = Avg. Velocity, P = Pressure, ρ = Fluid density, c = Wave celerity, θ = Pipe slope, f = Friction factor, D = Pipe diameter.

Linear Combination:

To convert these into ODEs, we combine them using an unknown multiplier λ :

$$L = L_1 + \lambda L_2 = 0.$$

$$L = \left[\frac{\partial V}{\partial x} (V + \lambda \rho c^2) + \frac{\partial V}{\partial t} \right] + \lambda \left[\frac{\partial P}{\partial x} \left(\frac{1}{\rho \lambda} + V \right) + \frac{\partial P}{\partial t} \right] + g \sin \theta + \frac{fV|V|}{2D} = 0$$

Recall the total derivatives for Velocity and Pressure: $\frac{dV}{dt} = \frac{\partial V}{\partial x} \frac{dx}{dt} + \frac{\partial V}{\partial t}$ and

$\frac{dP}{dt} = \frac{\partial P}{\partial x} \frac{dx}{dt} + \frac{\partial P}{\partial t}$. The bracketed terms in L become total derivatives if and only if:

$$\frac{dx}{dt} = V + \lambda \rho c^2 \quad \text{and} \quad \frac{dx}{dt} = \frac{1}{\rho \lambda} + V$$

Equating these conditions yields the exact value of the multiplier λ :

$$V + \lambda \rho c^2 = \frac{1}{\rho \lambda} + V \implies \lambda^2 = \frac{1}{\rho^2 c^2} \implies \lambda = \pm \frac{1}{\rho c}$$

Substituting λ back into the total derivative expression gives the characteristic curves $\frac{dx}{dt}$:

$$\frac{dx}{dt} = V \pm c \quad \dots (\text{Characteristic Lines})$$

And substituting λ into the combined equation L yields the simplified ODEs:

$$\frac{dV}{dt} \pm \frac{1}{\rho c} \frac{dP}{dt} + g \sin \theta + \frac{fV|V|}{2D} = 0 \quad \dots (\text{ODE form})$$

Finite Difference Solution of Characteristic Equations

To solve the ODEs numerically, we note that in pipe transients, $V \ll c$, thus $\frac{dx}{dt} \approx \pm c$. We integrate the ODEs along the positive (C^+) and negative (C^-) characteristics.

1. Integrating along C^+ (from point A to P):

Multiplying the ODE by $\rho c dt$ and integrating:

$$\rho c \int_A^P dV + \int_A^P dP + \int_A^P \rho c g \sin \theta dt + \int_A^P \rho c dt \frac{fV|V|}{2D} = 0$$

Since $dx = c dt$ along C^+ , we replace $c dt$ with dx . Integrating yields the discrete form:

$$\rho c (V_P - V_A) + (P_P - P_A) + \rho g \sin \theta \Delta x + \frac{\rho \Delta x f V_A |V_A|}{2D} = 0$$

2. Converting Velocity & Pressure to Discharge & Head:

Using the piezometric head relation $P = \rho g(H - Z)$, we evaluate the pressure difference:

$$P_P - P_A = \rho g(H_P - H_A) - \rho g(Z_P - Z_A)$$

Substituting $Z_P - Z_A = \sin \theta \Delta x$ (elevation change), the gravity term exactly cancels out:

$$P_P - P_A = \rho g(H_P - H_A) - \rho g \sin \theta \Delta x$$

Substituting this back and replacing velocity $V = Q/A_{area}$:

$$\frac{\rho c}{A_{area}} (Q_P - Q_A) + \rho g(H_P - H_A) + \frac{\rho \Delta x f Q_A |Q_A|}{2D A_{area}^2} = 0$$

Dividing the entire equation by ρg and rearranging to isolate H_P , we get the final explicit finite difference equations:

Along positive characteristic C^+ (A to P):

$$H_P = H_A - B(Q_P - Q_A) - R Q_A |Q_A|$$

Along negative characteristic C^- (B to P):

$$H_P = H_B + B(Q_P - Q_B) + RQ_B|Q_B|$$

Where the pipeline constants are defined as the **impedance coefficient** $B = \frac{c}{gA_{area}}$ and the **resistance coefficient** $R = \frac{f\Delta x}{2gDA_{area}^2}$.



Exam Questions Addressed:

- Derive a numerical solution for evaluating water hammer problem (characteristics on diagonal vs not on diagonal). [2079 Shrawan]
- Write an algorithm for simulation of water hammer process using MOC. [2072 Ashwin, 2070 Bhadra]
- Explain the criteria for the stability of the MOC solution. [2073 Bhadra]

6.5 Algorithm for Simulating Water Hammer via MOC

Case A: Rectangular Grid (Characteristics on Diagonal)

When the time step is deliberately chosen as $\Delta t = \Delta x/c$, the characteristic lines pass exactly through the pre-defined grid nodes (Courant Number = 1).

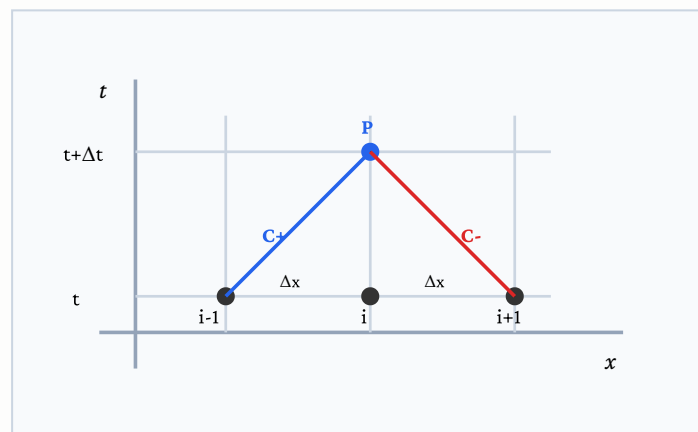


Figure: Characteristics passing exactly through nodes (Diagonal Grid)

Algorithm Steps:

1. Divide the pipe of length L into N reaches: $\Delta x = L/N$. Define time step $\Delta t = \Delta x/c$.
2. Compute pipeline constants: $B = \frac{c}{gA_{area}}$ and $R = \frac{f\Delta x}{2gDA_{area}^2}$.
3. Establish initial conditions (steady state): Know H and Q for all nodes $i = 1$ to $N + 1$ at $t = 0$.
4. Advance the time step to $t = t + \Delta t$.

5. For all interior nodes ($i = 2$ to N), evaluate the known constants from the previous time step:

$$C_P = H_{i-1} + Q_{i-1}(B - R|Q_{i-1}|)$$

$$C_M = H_{i+1} - Q_{i+1}(B - R|Q_{i+1}|)$$

6. Solve simultaneously for the interior unknowns:

$$H_{Pi} = \frac{C_P + C_M}{2} \text{ and } Q_{Pi} = \frac{C_P - H_{Pi}}{B}$$

7. For boundary nodes ($i = 1$ and $i = N + 1$), use the single applicable characteristic equation alongside the specific boundary condition (e.g., constant reservoir head or valve closure relation) to solve for local H and Q .

8. Repeat steps 4 through 7 until the final simulation time is reached.

Case B: Specified Time Intervals (Characteristics Not on Diagonal)

When Δt is independently specified such that $\Delta t < \Delta x/c$, the characteristic lines originating from point P fall *between* the established grid nodes at points L and R.

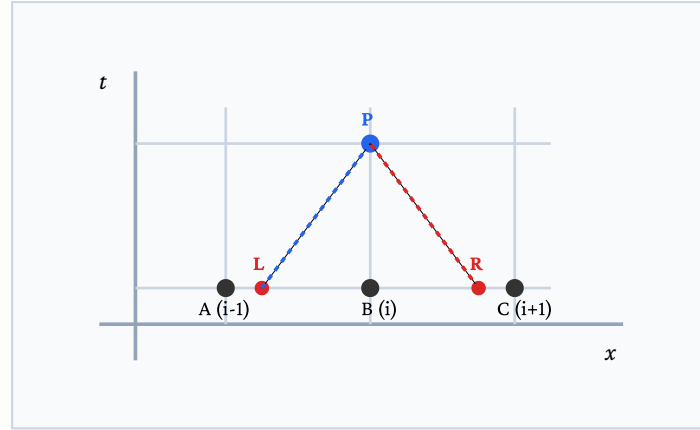


Figure: Characteristics NOT on diagonal (requires interpolation at L and R)

In this scenario, the values of Head (H) and Discharge (Q) at the foot of the characteristics (points L and R) are not explicitly known and must be found using linear interpolation from adjacent grid nodes A, B, and C.

Interpolation factor: $\theta = \frac{c\Delta t}{\Delta x}$

For Point L (between node i-1 and i):

$$Q_L = Q_i - \theta(Q_i - Q_{i-1}) \quad \text{and} \quad H_L = H_i - \theta(H_i - H_{i-1})$$

For Point R (between node i and i+1):

$$Q_R = Q_i - \theta(Q_i - Q_{i+1}) \quad \text{and} \quad H_R = H_i - \theta(H_i - H_{i+1})$$

Once H_L, Q_L, H_R, Q_R are obtained, they replace H_A, Q_A, H_B, Q_B in the standard C^+ and C^- compatibility equations. Furthermore, the resistance coefficient R must be scaled down relative to the characteristic length by substituting $c\Delta t$ in place of Δx .

Criteria for Stability of MOC

The stability of the explicit MOC numerical solution is governed by the **Courant Condition**.

$$\text{Courant Number } (C_r) = \frac{c\Delta t}{\Delta x} \leq 1$$

Physical Meaning: The characteristic paths represent the absolute speed at which physical pressure waves travel through the pipe. If $C_r > 1$, the characteristics originating from P will fall completely *outside* the known grid interval of $[i-1, i+1]$. This forces the numerical scheme to use spatial extrapolation rather than interpolation, which inherently amplifies truncation errors and causes the mathematical solution to rapidly become unstable and blow up.